HR Analytics Dashboard

Project Overview

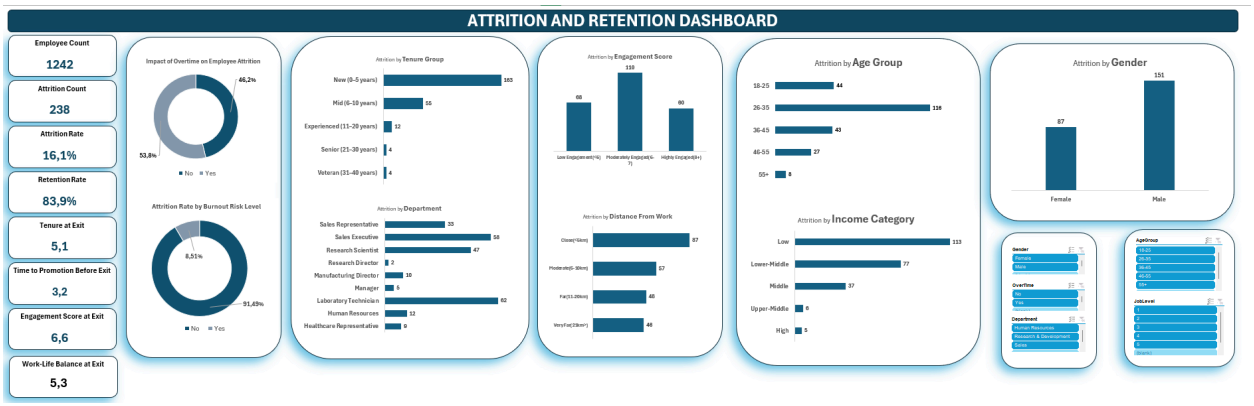
🎯 Project Objective

The objective of this project was to design and develop interactive HR dashboards that provide actionable insights into workforce dynamics. Using Excel and Power Query, I focused on analyzing key areas such as employee attrition, tenure distribution, and compensation trends. The dashboards aim to help HR teams and management make data-driven decisions, identify workforce challenges, and support strategic planning.

Data Preparation & Transformation

- **Tools Used:** Power Query (Excel).
- **Cleaning Steps:** Removed duplicates, handled missing values, and ensured consistency in data types (e.g., salary fields captured correctly, numeric vs text issues resolved).
- **Feature Engineering & Transformations:**

- Converted *Job Satisfaction* into a **/10 rating** to align with the *Engagement Score* metric.
- Grouped **distance from work** into categorical bands (e.g., short, medium, long commute) to simplify analysis.
- Created **income categories** to group employees into salary bands for easier comparisons.
- Converted **salary values** from USD to ZAR for local context. This was not a strict 1:1 exchange rate but adjusted to approximate current market salary levels.
- Grouped **tenure** into ranges (e.g., *0–1 years, 2–5 years, 6–10 years, 10+ years*) to better analyze retention patterns by employee experience.



The key figures in this dashboard show an **Attrition Rate of 16.1%** and a **Retention Rate of 83.9%** from a total workforce of **1,242 employees**. On average, employees leave after **5.1 years**, suggesting they typically spend a significant period with the company before exiting. The data also highlights a higher attrition among males (**151**) compared to females (**87**).

Trends and Patterns:

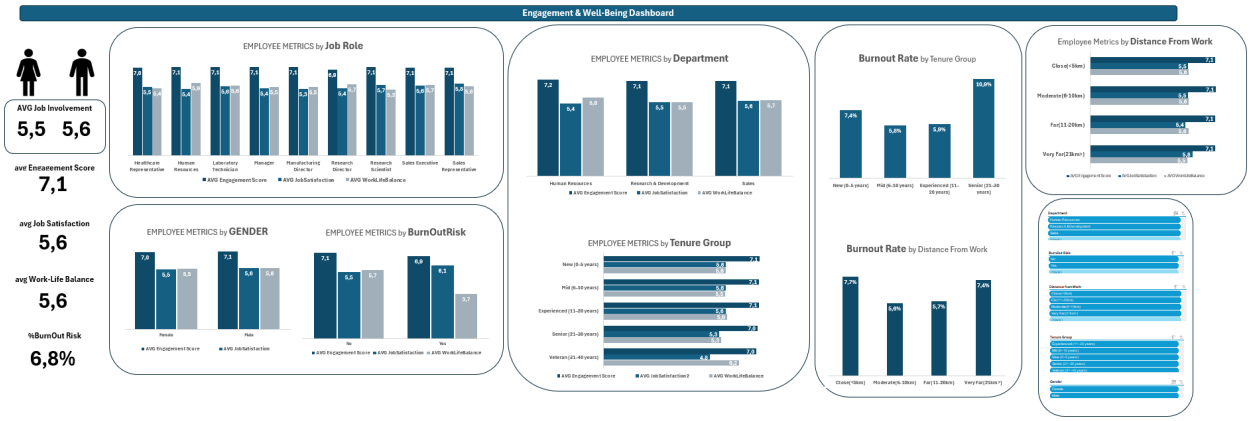
- **Age and Tenure:** The highest turnover is among employees aged **18–35**, with **160 exits**. This aligns with tenure data, showing the greatest attrition in the

New (0-5 years) and Mid-level (6–10 years) groups. This indicates that many new-career and mid-career professionals are leaving.

- **Departmental Attrition:** Roles with the highest exits are **Sales Executive (58)** and **Laboratory Technician (62)**, pointing to critical areas that need attention.
- **Overtime and Burnout:** More than half (**53.8%**) of those who left worked overtime, yet a large majority (**91.4%**) were not classified as at risk of burnout. This implies that while overtime may play a role, it isn't the main driver behind most resignations.
- **Work-Life Balance:** The average Work-Life Balance at exit is **5.3**, indicating that employees who leave the company report a relatively low score in this area at the time of their departure.

Implication:

The data suggests that the company is facing notable attrition challenges, particularly among younger and mid-career employees, with the highest exits in critical roles such as **Sales Executives** and **Laboratory Technicians**. Although overtime contributes to turnover, it is not the primary factor, as most employees leaving are not classified as burnout risks. Instead, the **low work-life balance score (5.3)** points to potential dissatisfaction with overall employee experience and lifestyle alignment. Combined with higher male attrition rates, these trends indicate that the company may need to strengthen **career development opportunities, retention strategies in high-exit roles, and work-life balance initiatives** to address the underlying drivers of attrition.



Key Metrics and Analysis:

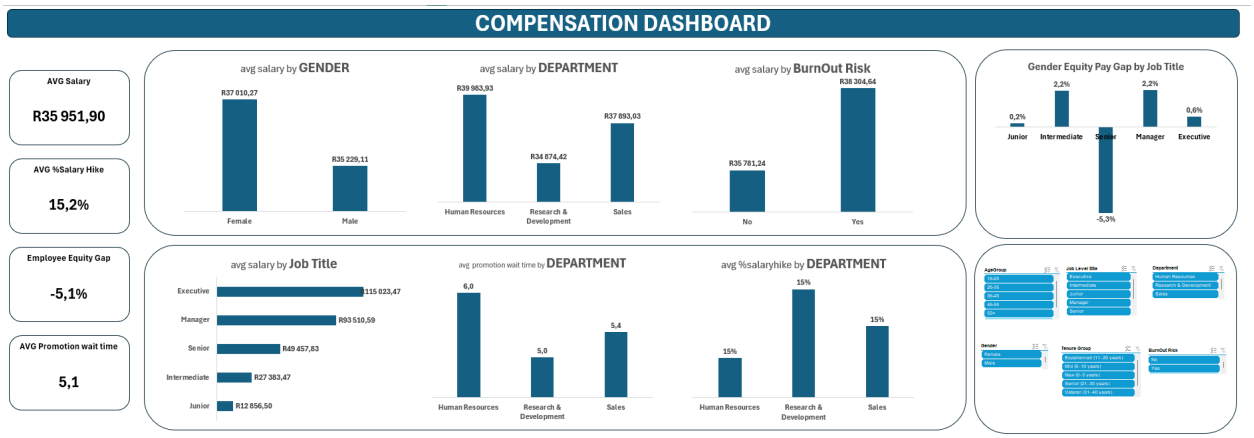
This dashboard highlights the qualitative side of employee experience. The main indicators show an **average Engagement Score of 7.1**, an **average Job Satisfaction Score of 5.6**, and a **low average Burnout Risk of 6.8%**.

Trends and Patterns:

- **Low Job Satisfaction:** A key trend is the notably low Job Satisfaction score (5.6), which lags behind the Engagement Score (7.1). This indicates that while employees are engaged in their work, they are not necessarily satisfied with their roles.
- **Departmental Differences:** The **HR department** reports the lowest scores across engagement, job satisfaction, and work-life balance. In contrast, the **Sales department** shows high engagement but struggles with low job satisfaction and work-life balance, along with a heightened burnout risk.
- **Gender Differences:** Female employees score slightly higher in engagement and work-life balance compared to males, but report lower job satisfaction.
- **Impact of Experience:** Engagement scores are highest among veteran and long-tenured employees, while those with fewer than 5 years of experience report the lowest scores suggesting that engagement grows with tenure.
- **Burnout by Tenure and Distance:** Senior employees (21–30 years of tenure) face the highest burnout risk at **10.9%**. Additionally, those living **very close (<5km)** or **very far (>21km)** from work have elevated burnout risks (7.7% and 7.4% respectively), showing that both short and long commutes can contribute to stress.

Implications:

The gap between high engagement and low job satisfaction is a critical insight. It suggests that while employees are active and contributing, their core needs—such as fair pay, career development, and better work-life balance—may not be fully met. The challenges within **HR and Sales** departments call for urgent attention, while the burnout trends tied to **tenure and commute distance** highlight specific groups that may need targeted support to improve overall well-being.



Key Metrics and Analysis:

This dashboard offers a clear view of salary and compensation insights. The **average salary is R35,951.90**, with a notable **average salary hike of 15.2%**. The **Employee Equity Gap of -5.1%** suggests a pay imbalance that favors female employees.

Trends and Patterns:

- **Gender Pay Gap:** Female employees earn a significantly higher average salary (**R37,910.37**) compared to males (**R35,229.11**). This aligns with the negative equity gap, showing a pay disparity in favor of women.
- **Job Title and Salary:** As expected, salaries rise with seniority, ranging from **Junior (R12,856.50)** to **Executive (R115,103.47)**. The **Sales department** shows the largest average hike at **15%**.
- **Burnout and Compensation:** Employees with a high burnout risk earn more on average (**R38,304.64**) than those at lower risk, suggesting that higher pay may come with added pressure and greater burnout potential.

Implications:

The compensation data indicates a potential link between **higher pay and burnout**, particularly in **Sales**. The gender pay gap, while showing a disparity in favor of women, underscores the need for a **deeper, more nuanced review of pay equity** across roles and tenure to fully understand its long-term impact.

Synthesizing Insights Across Dashboards

The data from these three dashboards is deeply interconnected, creating a holistic view of the employee experience.

Attrition and Compensation:

High attrition in the **Sales department** (from the Attrition Dashboard) aligns with the **highest average salary hikes** in the same department (from the Compensation Dashboard). This shows that while the company offers competitive raises, it may not be enough to retain staff. The **high burnout rates tied to higher pay** also connect directly to the turnover seen in high-pressure Sales roles. Additionally, the corrected gender pay gap data provides an important link: the higher male attrition rate may be influenced by the **negative equity gap favoring females**.

Engagement and Attrition:

The consistently **low Job Satisfaction scores** (from the Engagement Dashboard) are likely a major driver of attrition. While employees may stay engaged in daily work, dissatisfaction with their roles, pay, or work-life balance appears to push them to leave. This is most evident in **Sales and Research**, where attrition is high despite engagement also being high.

Promotion and Attrition:

The **average promotion wait time of 3.2 years** for employees who leave (from the Attrition Dashboard) ties directly to the **low Job Satisfaction scores** (from the Engagement Dashboard). The promotion process may not meet employee expectations for career growth, leading new and mid-career professionals, the group with the highest attrition to move on after years of service.

Work-Life Balance and Attrition:

The **low average Work-Life Balance score of 5.3** at exit strongly signals it as a key factor behind turnover. This metric, alongside job satisfaction and career progression, should be a priority for addressing retention challenges. New data on **burnout by commute distance** adds another layer: employees with either very short or very long commutes tend to have poor work-life balance, raising their likelihood of leaving.

Overarching Trends:

The overall analysis makes one thing clear: it’s crucial to go beyond surface-level fixes. There is a **complex interplay between pay, job satisfaction, career growth,**

and work-life balance. Simply raising salaries won’t resolve attrition if deeper issues like slow promotions, burnout, and dissatisfaction remain unaddressed, especially in critical departments like Sales. Targeted solutions, such as adopting a **hybrid work model**, could directly ease work-life balance challenges for long-commute employees and help improve retention.